

Frequency-Domain Deblurring

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Implementing and comparing Fourier-domain restoration methods against spatial-domain baselines for deblurring

Image source

Clean reference images are taken from **CBSD68** (68 colour natural images, 321×481), obtained from the benchmark collection at https://github.com/majedelhelo/denoising_datasets. CBSD68 is the standard colour image-denoising benchmark, derived from the Berkeley Segmentation Dataset (BSDS500), so the PSNR and SSIM values reported here are directly comparable with published restoration results. The collecting repository states that the datasets are gathered for convenience, with no ownership claimed.

Degradation model

$$g(x, y) = h(x, y) * f(x, y) + n(x, y) \quad (1)$$

$$G(u, v) = H(u, v) F(u, v) + N(u, v) \quad (2)$$

f = sharp image, h = blur kernel (PSF), n = additive noise, g = observed blurry image; capitals denote the corresponding 2-D discrete Fourier transforms, and $\hat{F}$ denotes the restored estimate of F .

Inverse filter

$$\hat{F}(u, v) = \frac{G(u, v)}{H_\varepsilon(u, v)}, \quad H_\varepsilon = \begin{cases} H, & |H| \geq \varepsilon, \\ \varepsilon, & |H| < \varepsilon, \end{cases} \quad \varepsilon = 10^{-2} \quad (3)$$

Wiener filter

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + K} G(u, v), \quad K = 10^{-2} \quad (4)$$

H^* = complex conjugate; K approximates the noise-to-signal power ratio S_n/S_f .